

AI MENTAL HEALTH

ANA – 2 (WEEK-3)

Drop-off & DNA Rate Analysis

**Dataset Used: Psychological Therapies Annual Report
(IAPT Dataset)**

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1. Introduction

Mental health services often face challenges related to patient disengagement during treatment. Patients may fail to attend therapy sessions, discontinue treatment early, or drop out before completing a therapeutic course. These issues reduce treatment effectiveness and increase pressure on healthcare systems.

This project analyses:

- Did Not Attend (DNA) rates
- Treatment drop-off rates
- Patient progression through treatment stages
- Regional variations in disengagement

The analysis was performed using the NHS Psychological Therapies annual report dataset.

2. Objectives

The main objectives of this analysis were:

1. Compute estimated DNA rates

2. Analyze treatment drop-off patterns
3. Measure treatment completion rates
4. Build a treatment funnel chart
5. Compare DNA rates across regions
6. Export analytical summaries as CSV files

3. Dataset Description

Dataset Used:

Psychological Therapies Annual Report (IAPT Dataset)

The dataset contains mental health service statistics including:

- Referrals received
- Patients accessing services
- Treatment sessions attended
- Completed treatment cases
- Early treatment endings
- Regional/provider information

4. Data Preprocessing

Several preprocessing steps were performed before analysis.

4.1 Data Cleaning

- Removed non-total breakdown records
- Converted numeric columns into usable numerical format
- Handled missing and invalid values

- Removed formatting symbols such as *

4.2 Feature Selection

The following variables were selected for analysis:

- Count_ReferralsReceived
- Count_AccessingServices
- Count_SecondTreatment
- Count_FinishedCourseTreatment
- Count_EndedBeforeTreatment
- Count_EndedAfterOneTreatment

5. Methodology

5.1 DNA Rate Calculation

The estimated DNA rate was calculated using:

$$[\text{DNA RATE} = (\text{Ended After One Treatment} / \text{Accessing Services}) * 100]$$

These metric estimates disengagement after minimal therapeutic interaction.

5.2 Drop-off Before Treatment

This metric identifies patients who left the service before treatment began.

$$[\text{Dropoff Before Treatment} = (\text{Ended Before Treatment} / \text{Referrals Received}) * 100]$$

5.3 Drop-off After First Session

This measures patients discontinuing treatment after initial engagement.

$$[\text{Dropoff After First Session} = (\text{Ended After One Treatment} / \text{Second Treatment}) * 100]$$

100

]

5.4 Treatment Completion Rate

The treatment completion rate was computed as:

[

Completion Rate = (Finished Course Treatment/ Accessing Services)* 100

]

6. Funnel Analysis

A funnel analysis was created to visualize patient movement through treatment stages.

The funnel stages included:

1. Referrals Received
2. Accessing Services
3. Second Treatment Session
4. Finished Treatment

The funnel chart demonstrates progressive patient reduction across treatment stages.

7. Regional DNA Rate Analysis

Regional comparison analysis was performed by grouping records according to regional categories and calculating average DNA rates.

The visualization highlighted:

- Regions with higher disengagement

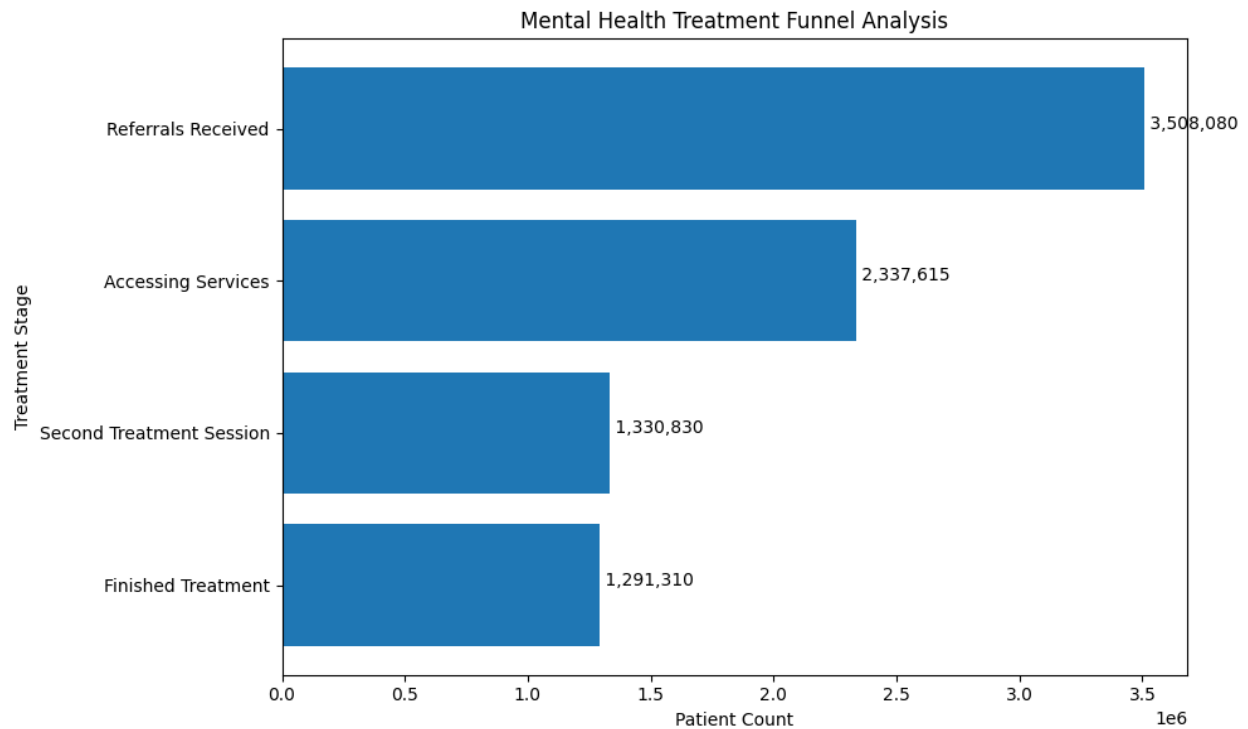
- Regions with comparatively better treatment retention
- Variations in patient attendance patterns

8. Visualizations Generated

The following visualizations were produced:

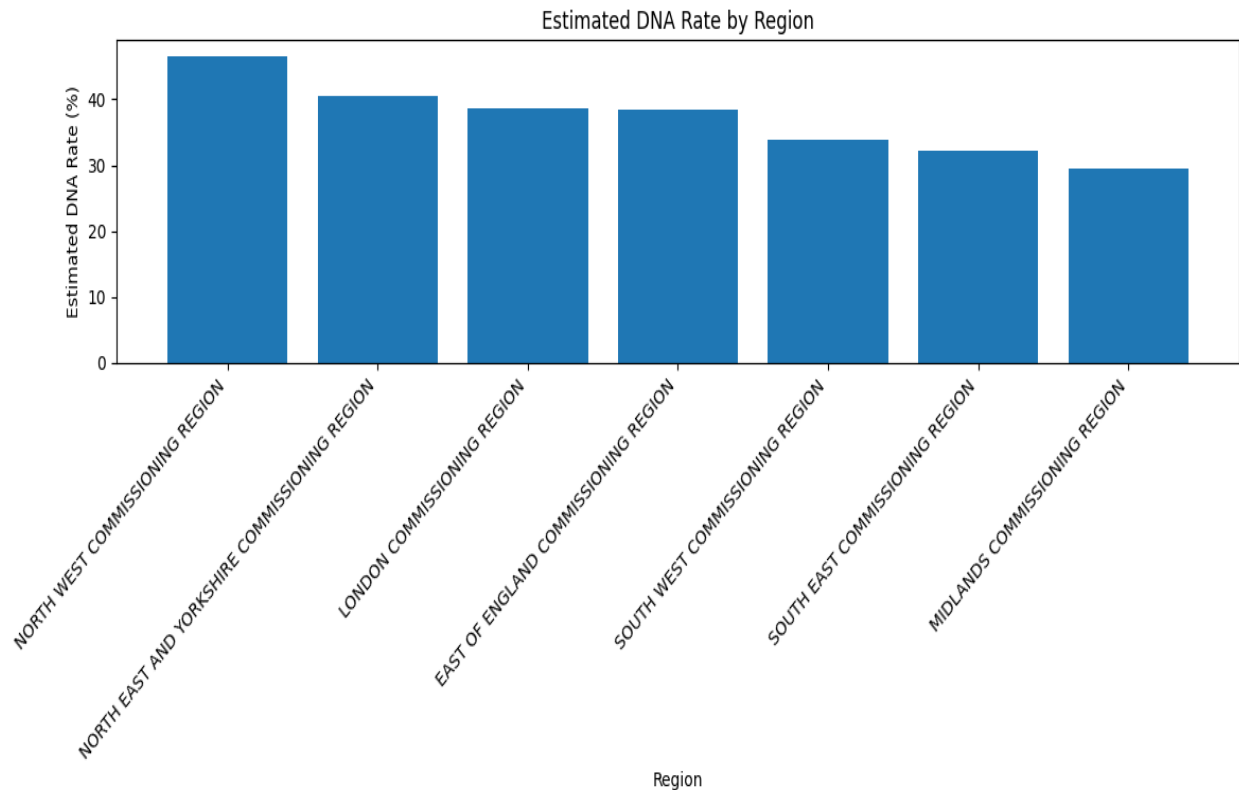
8.1 Funnel Chart

Displays patient progression through treatment stages.



8.2 Estimated DNA Rate by Region

A bar chart comparing regional DNA percentages.



9. CSV Outputs

The following CSV files were generated:

File Name	Description
DNA_Dropoff_Analysis.csv	DNA and drop-off rate summary
Treatment_Funnel_Summary.csv	Funnel stage totals

10. Key Findings

10.1 Patient Attrition

The funnel analysis showed a noticeable reduction in patient numbers between referral and treatment completion stages.

10.2 Early Treatment Disengagement

A significant proportion of patients disengaged after the first treatment interaction.

10.3 Regional Variability

Some regions exhibited substantially higher estimated DNA rates, indicating possible differences in:

- Service accessibility
- Waiting times
- Patient engagement
- Resource allocation

10.4 Treatment Completion

Completion rates indicated that many patients successfully completed treatment, although substantial attrition remained across earlier stages.

11. Conclusion

This analysis successfully identified treatment disengagement patterns using NHS mental health therapy data.

The study demonstrated how funnel analysis and DNA rate estimation can help healthcare organizations:

- Identify service bottlenecks
- Improve patient engagement strategies
- Reduce treatment attrition
- Enhance mental health service effectiveness

Future improvements could include:

- Time-series DNA analysis
- Age and gender-based disengagement analysis

- Predictive modelling for dropout risk
- Geographic hotspot mapping